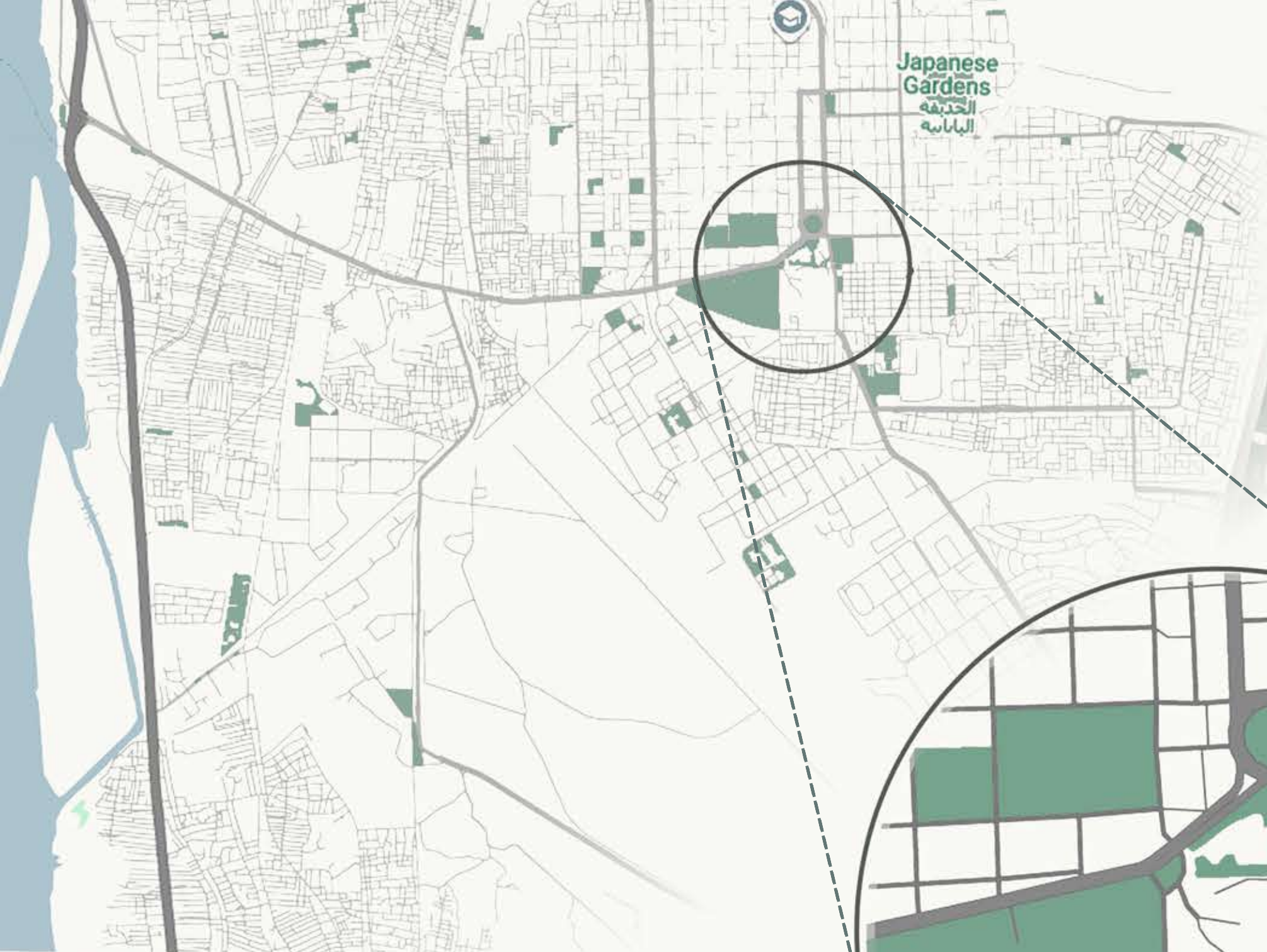




MSA UNIVERSITY
جامعة أكتوبر للعلوم الحديثة والآداب



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ASE 5532 FALL 25 Thermal Comfort in Architecture

This project investigates thermal comfort in architecture through the climatic analysis and environmental simulation of a spa accommodation building designed by us in Architectural Design VI course, in Helwan, Cairo. Using Climate Consultant and DesignBuilder, the study follows a four-phase methodology: analyzing Cairo's climate, examining a climate-responsive local case study, evaluating the original design as a base case, and developing an optimized design scenario.

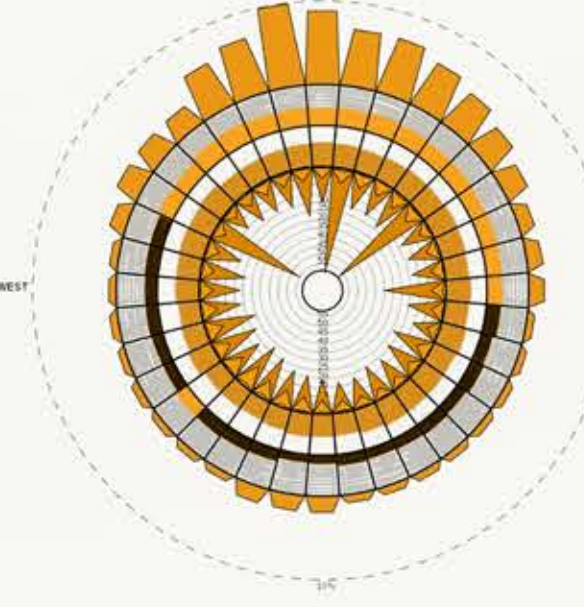
The project addresses RIBA Theme & Value 1 (Health and Life Safety) by improving indoor thermal comfort, Theme & Value 3 (Structures, Construction, and Resources) through envelope performance and U-value optimization, and Theme & Value 4 (Histories, Theories, and Methodologies) by applying climate-responsive design principles supported by simulation-based analysis.



PHASE 1: Cairo's Climatic Analysis

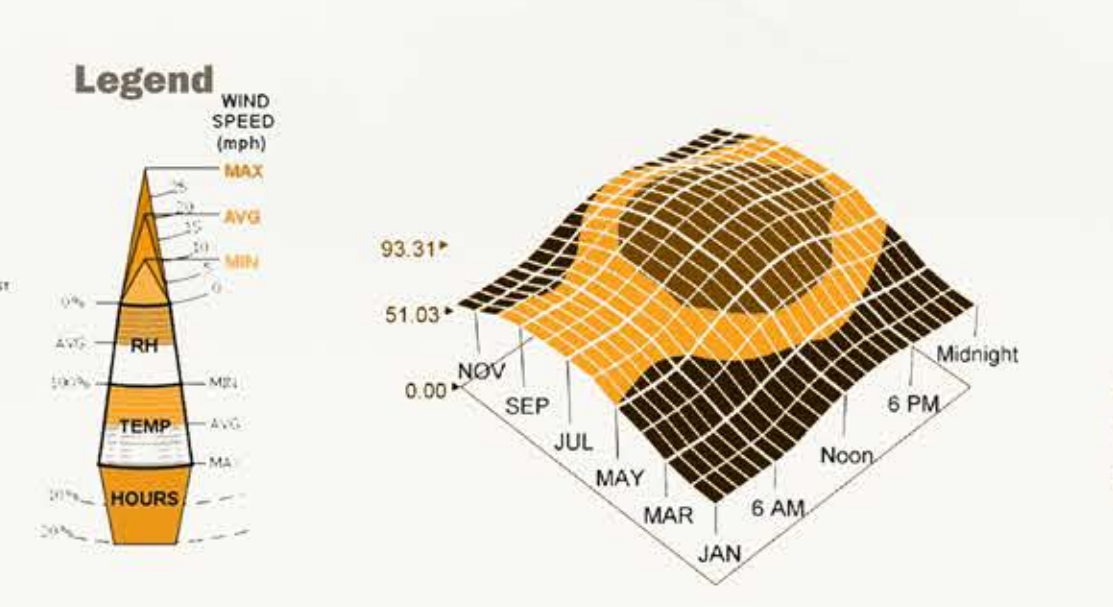
The first phase focuses on understanding the climatic conditions of Cairo using Climate Consultant, including temperature ranges, solar radiation, wind patterns, humidity levels, and thermal comfort zones. This phase establishes the environmental constraints and passive design opportunities that directly influence building performance. By identifying climate-responsive strategies suitable for a hot arid context, this phase forms the environmental foundation of the design process.

Wind Wheel



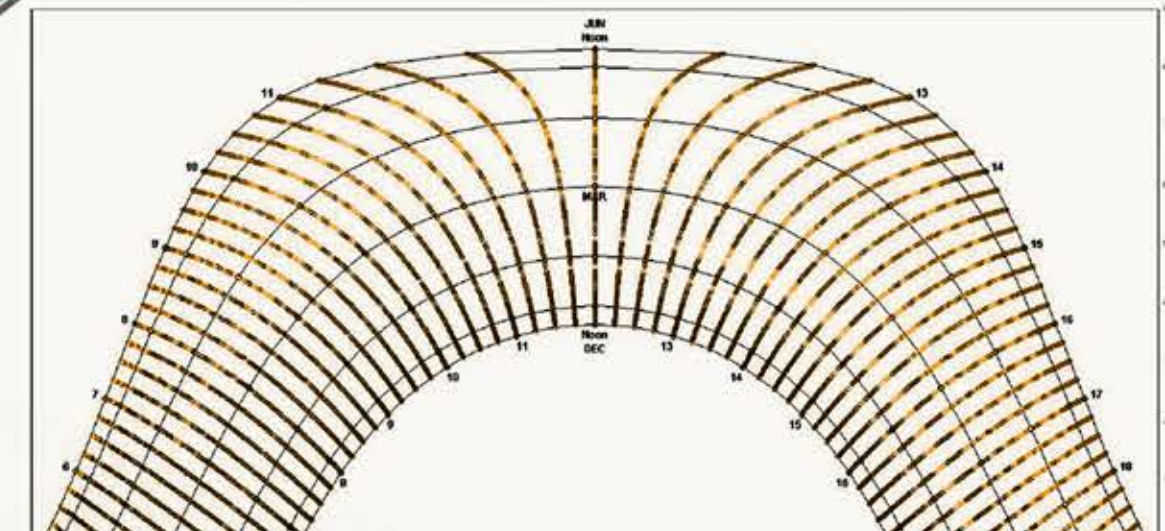
This diagram illustrates the prevailing wind directions and frequencies in Cairo, identifying the primary orientations for natural ventilation.

Dry-Bulb Temperature



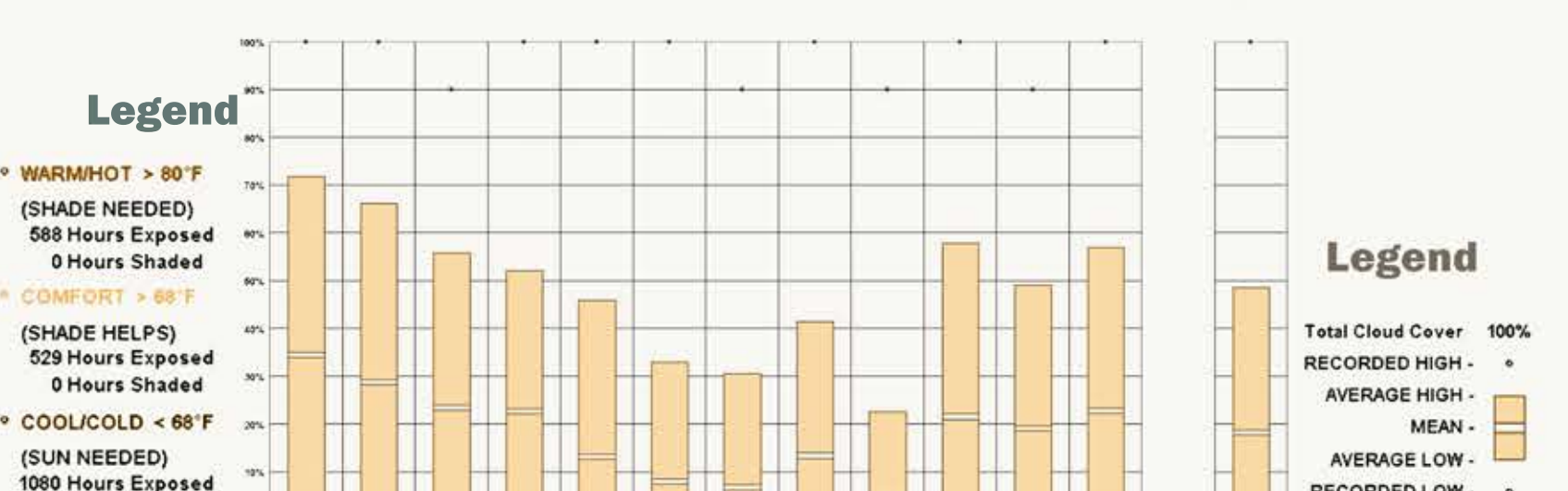
This surface plot visualizes diurnal and seasonal temperature fluctuations, highlighting peak heat periods across the calendar year.

Sun Shading Chart



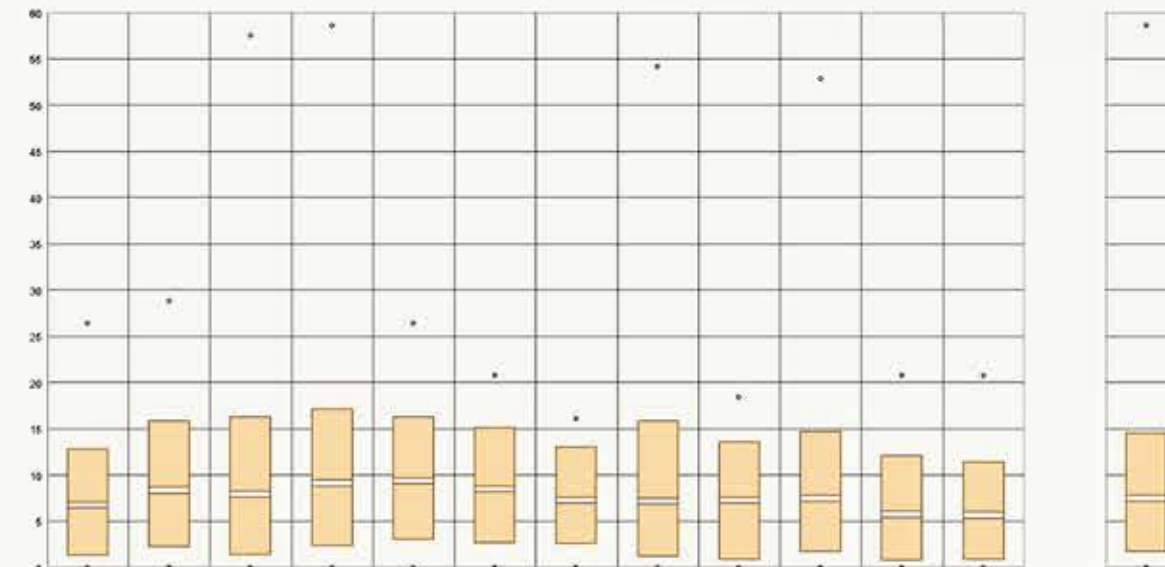
This stereographic projection maps solar paths throughout the year to determine the specific angles required for effective building shading.

Sky Cover Range



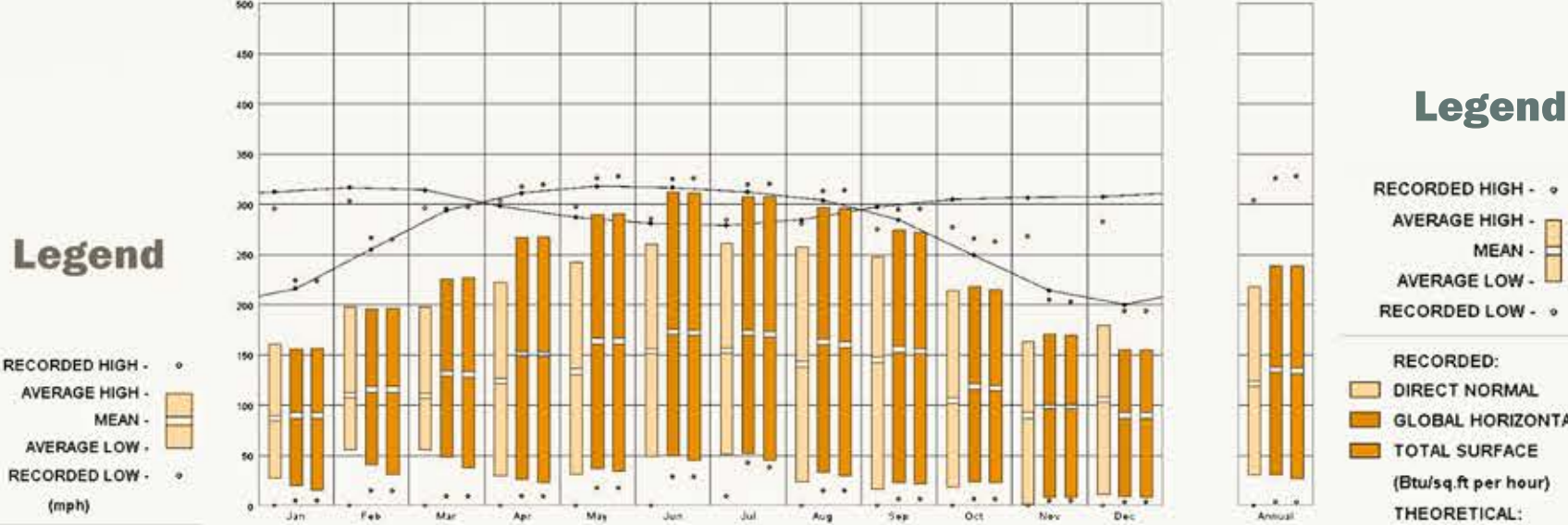
This chart tracks annual cloud cover patterns to assess the availability of daylight and the potential for passive solar gain.

Wind Velocity Range



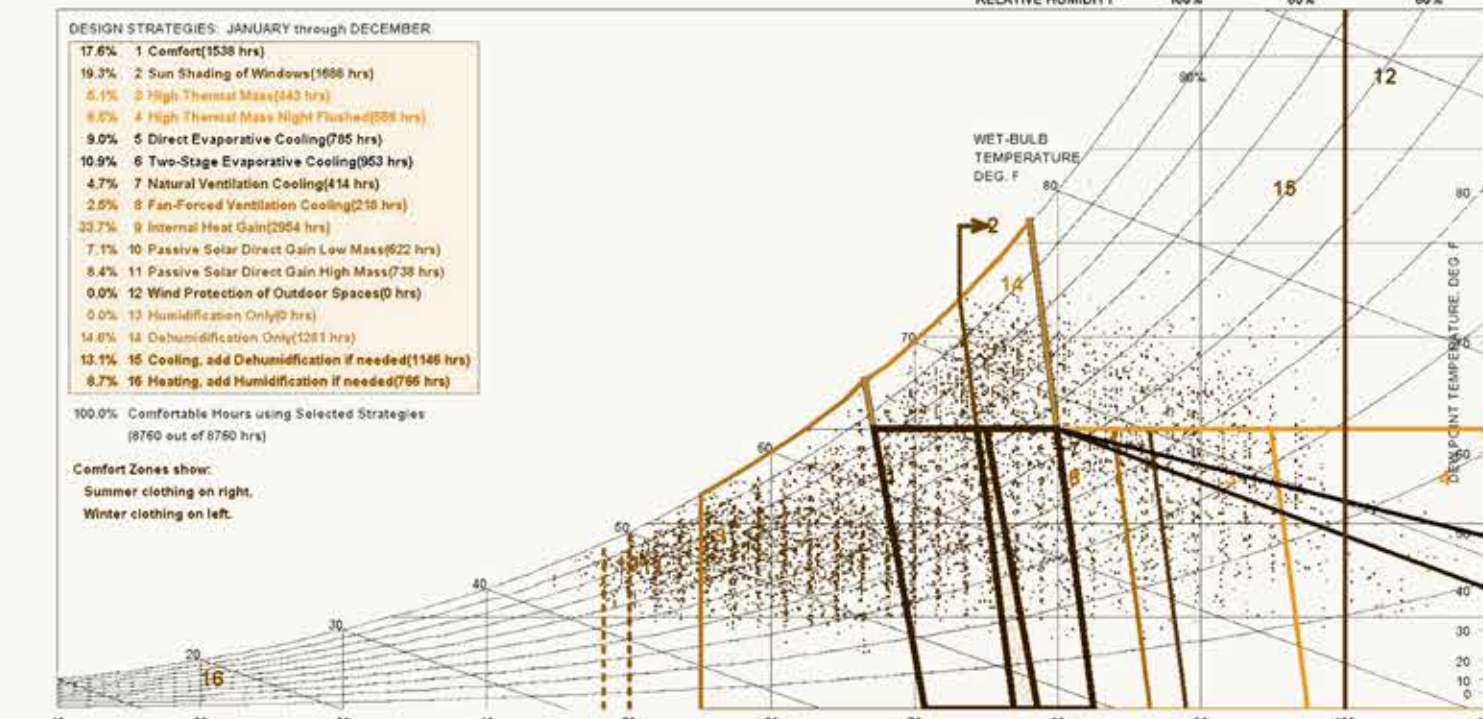
This graph displays monthly variations in wind speed, helping to evaluate the potential for consistent cross-ventilation.

Radiation Range



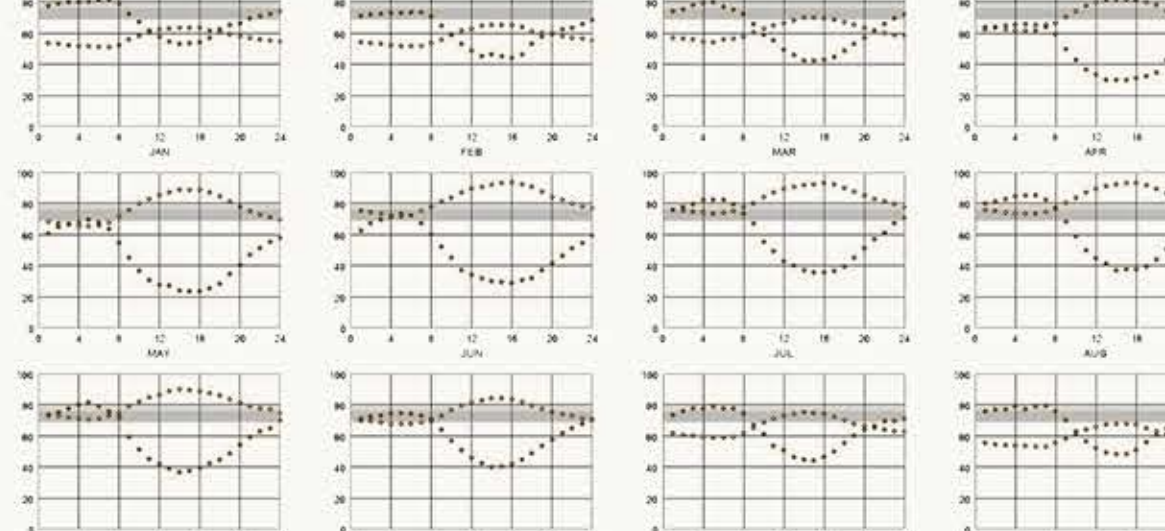
This chart quantifies the intensity of solar radiation received, distinguishing between direct and diffuse components for thermal load calculations.

Psychrometric Chart



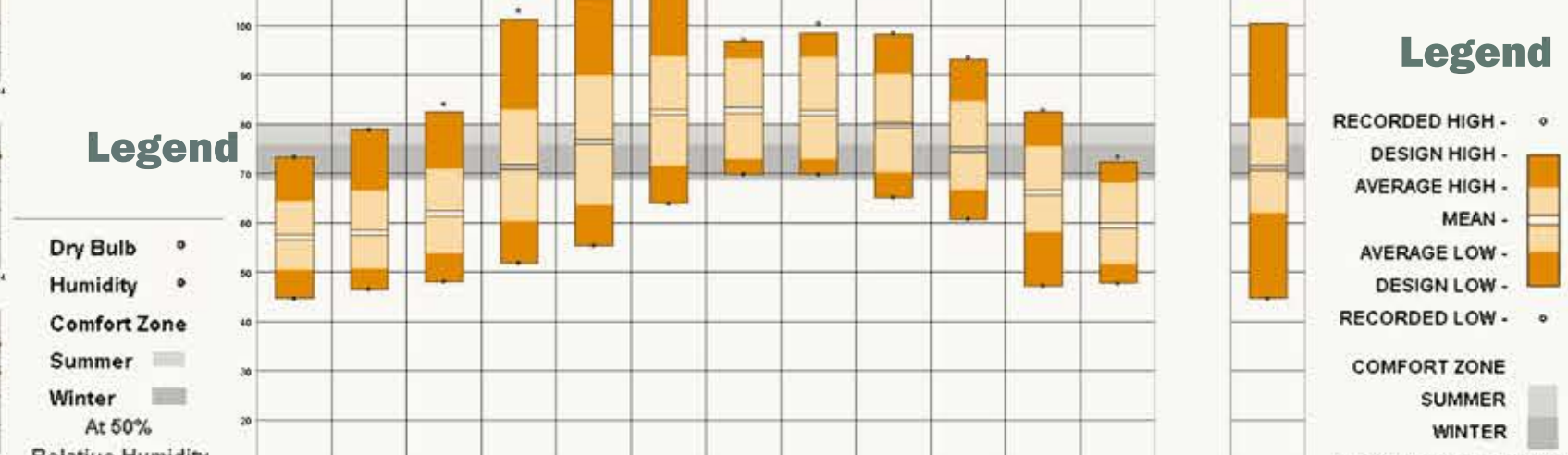
This plot maps the relationship between temperature and humidity to identify the most effective passive cooling strategies for the local climate.

Dry-Bulb X Relative Humidity



This matrix correlates temperature with moisture levels to pinpoint specific hours of the year that fall within the human comfort zone.

Temperature Range



This summary graph displays monthly highs, lows, and averages to establish the baseline thermal environment for the building envelope design.

PHASE 2: Case Study in Cairo

The second phase examines Bayt-Al Suhaymi in Cairo that demonstrates successful thermal comfort performance through passive and environmental design strategies. Bayt al-Suhaymi, or the House of Suhaymi, is a remarkably well preserved traditional Egyptian house and museum in the heart of Islamic Cairo, showcasing the opulent lifestyle and architectural genius of the Ottoman period.

The case study is analyzed in terms of orientation, envelope design, shading systems, material selection, and natural ventilation strategies. This phase translates theoretical climatic principles into practical architectural applications relevant to the local context.



Location: Darb al-Asfar, Al-Muizz Street, Cairo
Construction date: 1648 CE (Ottoman period), later expanded in 1796
Typology: Traditional Cairene residential house
Climate Context: Hot-dry climate (Cairo) with high diurnal temperature variation, strong solar exposure, and dusty environment.

Materiality & Structure

Thick Walls: Provide thermal mass, slowing heat transfer.

Marble & Water: In the qa'a (main hall), marble surfaces and a central fountain reflect and cool air.



Mashrabiyya (Screens): Wooden latticework screens provide shade while allowing airflow.



1.1 Microclimate: The courtyard, with plants and trees, creates a cooler, humid oasis, generating higher pressure than the hotter exterior.

Courtyard & Garden (Bustan):



1.2 Evaporative Cooling: Water features and greenery absorb solar heat and release moisture, cooling the air.

Spatial Zoning (Takhtabush & Maq'ad):

3.1 Takhtabush: A shaded, covered outdoor sitting area that facilitates cool air flow from the courtyard into the house.



2 Maq'ad (Sitting Room): Often north-facing to catch cool winds, serving as a comfortable evening gathering spot.

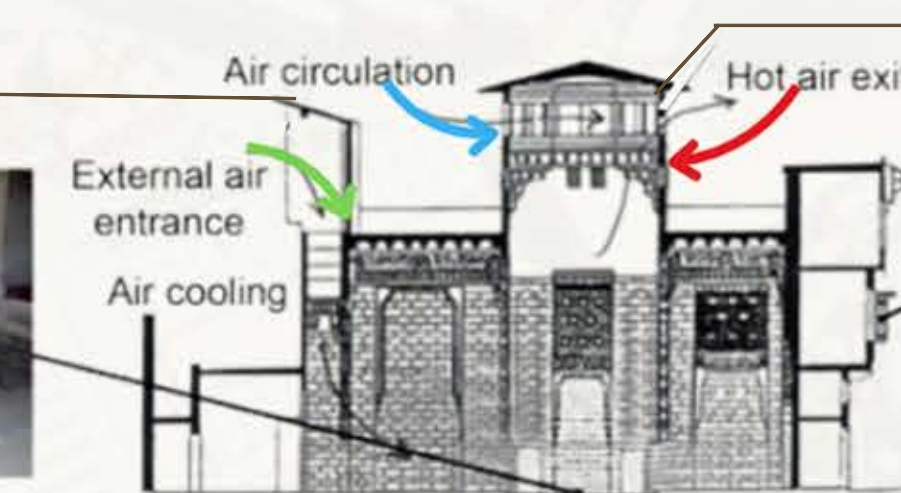
4.2 Fountains enhance evaporative cooling



Malqaf (Wind catcher)



Water Fountain



Mashrabiyya



Shokhshikha

Ventilation (Malqaf & Shokhshekha):

Vegetation & Water Features

4.1 Trees in the courtyard create shaded pockets and improve humidity.





PHASE 3: Base Case

Layers:
4 Layers: Plaster, Brick, Concrete Block, Gypsum

-Total Thickness:
0.325 m

U-Value:
1.475 W/m²K (High heat loss/gain)



The Base Case represents the building with a standard, uninsulated envelope (U-value of 1.475 W/m²K)

1. Temperature Profile: Indoor operative temperatures rise from approximately 20 C in January to a peak of 33-34 C in July and August

2. Solar Gains: External windows are a massive heat source, with solar gains peaking between 3,800 and 4,200 kWh/month during the summer

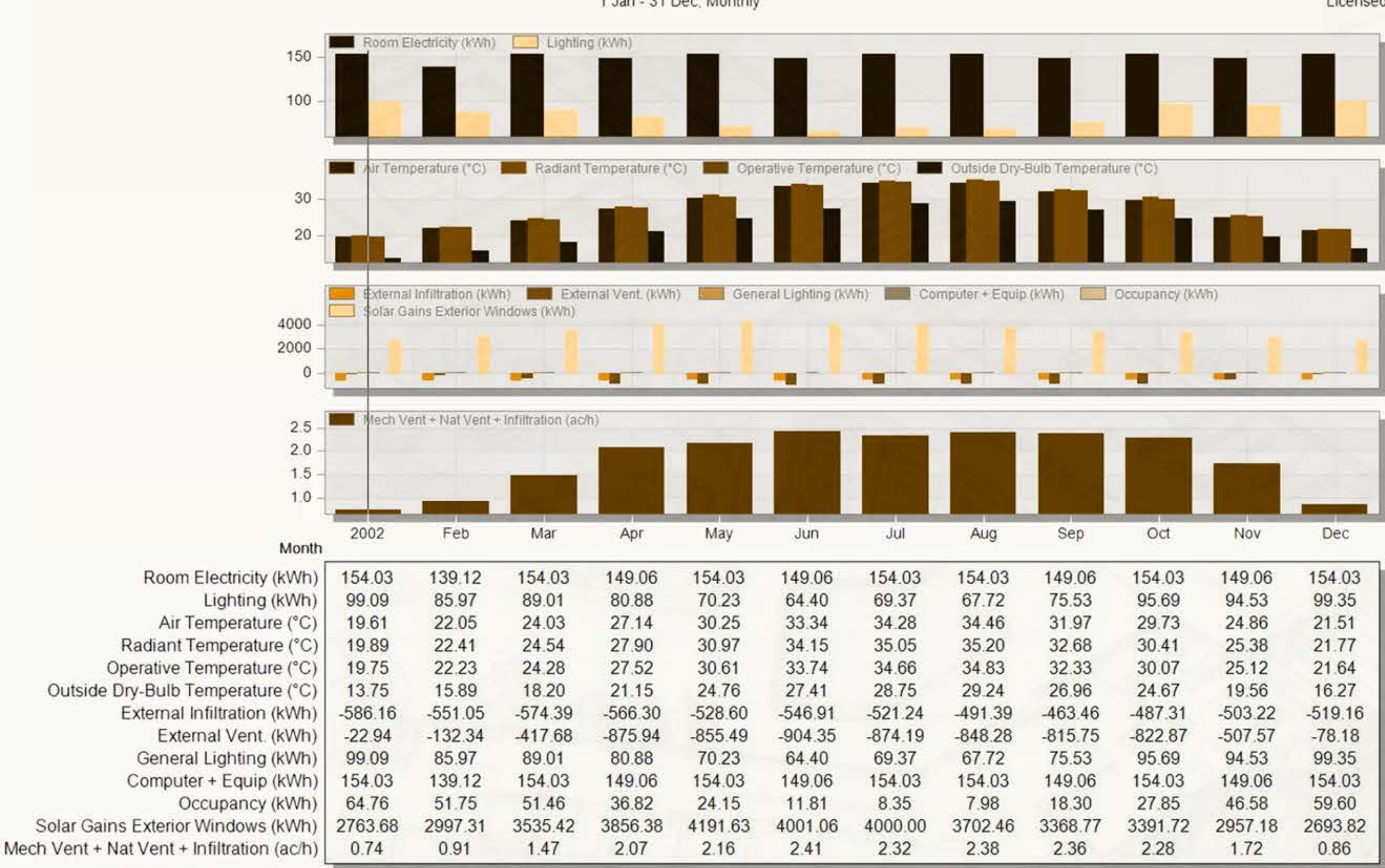
3. Energy Drivers: Room Electricity: Stays constant at roughly 154 kWh/month

4. Lighting: Varies seasonally, reaching its lowest point of 64.40 kWh in June due to natural daylight.

5. Ventilation: The combined rate (mechanical + natural + infiltration) increases from 0.7 ach in January to 2.4–2.6 ach in summer

6. Infiltration: Negative values in winter (up to -586 kWh) indicate significant heating losses during cold months.

Temperatures, Heat Gains and Energy Consumption - DESIGN 7, Building 1

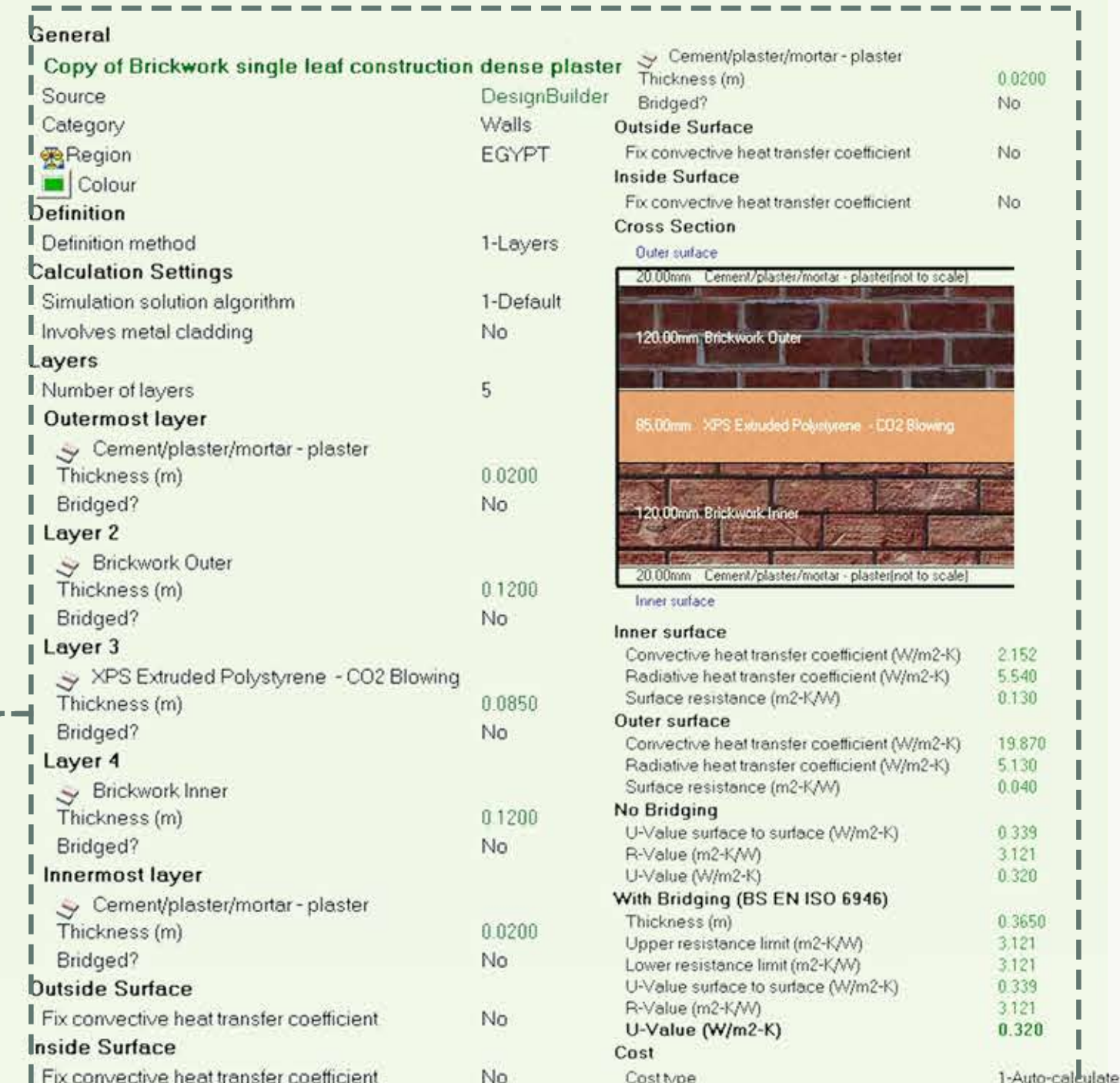


PHASE 4: New Case

Layers:
5 Layers: Plaster, Outer Brick, XPS Insulation (85mm), Inner Brick, Plaster

-Total Thickness:
0.365 m

U-Value:
0.320 W/m²K (High resistance)



The New Case features an insulated wall assembly (U-value of ~0.32 W/m²K) and reflects an active cooling/heating strategy

1. Energy Dominance: The profile is strictly cooling-dominated

2. Cooling electricity peaks in July at 2,324.99 kWh10

3. Heating Demand: Minimal and restricted to winter; peaks at 396.71 kWh in December (Gas)

4. Controlled Comfort: Operative Temperature: Is better regulated, peaking at 35.87°C in August

5. Air Temperature: Stays more stable, typically ranging between 21-15C and 26-12C for most of the year

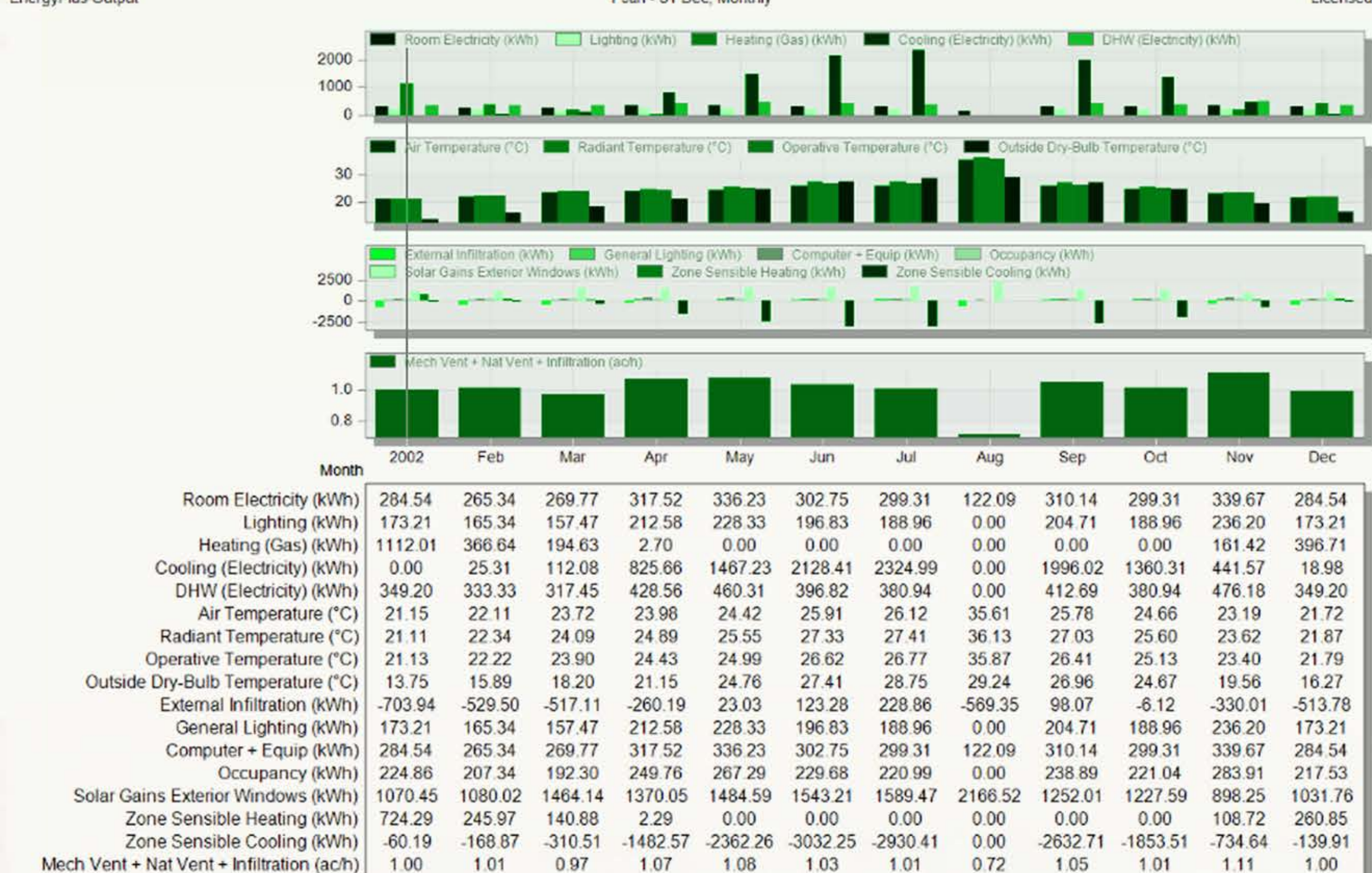
6. Solar Heat Reduction: Solar gains are significantly lower than the Base Case, peaking at 2,166.52 kWh in March and dropping to 1,484.59 kWh in June14

7. Internal Loads: Lighting and equipment loads remain consistent with the Base Case, showing these gains are independent of the envelope changes (e.g., equipment peaks at 339.67 kWh in November)

284.54

EnergyPlus Output

Temperatures, Heat Gains and Energy Consumption - DESIGN 7, Building 1



CONCLUSION

Key Improvements in the New Case:

1- Thermal Resistance: The addition of an 85mm XPS insulation layer significantly reduced the U-value from 1.475 to 0.320 W/m²K, effectively blocking excessive heat gain.

2- Energy Profile Shift: While the Base Case showed higher relative loads in winter (heating-dominated), the New Case highlights cooling as the primary driver, peaking between June and August due to the hot climate context.

3- Efficiency: The insulated envelope provides a much more stable foundation for energy efficiency, though passive cooling strategies (like shading) are still recommended to further reduce the summer cooling load

